



DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001-2015 Certified Institution)

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

INTERSHIP FINAL REVIEW (21INT82)

Web Development and Data Science for Process Automation at Toyota

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Guide: B G Mahesh, Manager

Organization: Toyota Kirloskar Motors Pvt. Ltd.

Internship Duration: Jan 2025 – July 2025



Objective & Scope

Objective:

- Build a robust, integrated web portal for automating workflows and approvals.
- Enable future AI/ML integration for Toyota employee services.

Scope:

- ☒ Data integration with internal databases & hierarchies
- ☒ Digital forms replacing paperwork
- ☒ Portal for AI/ML-based applications (internal use)
- ☒ Vehicle requisition automation
- ☒ Improved tracking, scheduling, feedback
- ☒ Reduced manual time (target: >100% improvement)



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Problem Statement / Research Question

Problem Statement:

- Manual vehicle requisition forms cause delays and inefficiencies
- No centralized tracking of approvals or requests

Research Question:

- How to design a web-based system to digitize and automate Toyota's vehicle requisition process
- eliminating paperwork, accelerating approvals, and enabling insights?



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Literature Review / Prior Work

- SharePoint and Microsoft Power Automate for workflows
- LDAP & Active Directory for access control
- Internal Toyota pilots on digital HR systems
- AI/ML in enterprise portals for predictive analytics



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Methodology / Tools Used / Technology Stack

- Requirement Analysis: Interviews with stakeholders
- Design: Wireframes in Figma, feedback iteration
- Development:
 - Frontend: JavaScript (jQuery, Bootstrap)
 - Backend: PHP (Laravel)
 - Database: MySQL + LDAP
 - Automation: SharePoint API
- Deployment: XAMPP, Azure VM
- AI/ML Ready APIs for future use



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Weekly Progress Summary

Phase	Week	Dates	Activities
Orientation & Research	Week 1	Jan 24 – Jan 30	Orientation to IT protocols, Toyota systems, compliance & digital landscape
	Week 2	Jan 31 – Feb 6	Studied internal tools (SharePoint, Outlook workflows), process mapping
Requirement Analysis & Design	Week 3	Feb 7 – Feb 13	Requirement gathering, stakeholder interviews, process flow documentation
	Week 4	Feb 14 – Feb 20	UI wireframing with Figma, finalized user roles and dashboard mockups
Backend Setup & Development	Week 5	Feb 21 – Feb 27	Set up LAMP stack, DB schema design, employee/vehicle/request models
	Week 6	Feb 28 – Mar 6	Implemented backend logic, DB normalization, and custom SQL query support
Frontend Integration	Week 7	Mar 7 – Mar 13	Developed request form UI, admin panel, and approval pages with HTML/CSS/JS
	Week 8	Mar 14 – Mar 20	Connected frontend with backend; real-time request tracking integration
Authentication & Role Setup	Week 9	Mar 21 – Mar 27	LDAP integration (Active Directory login), role-based access control
	Week 10	Mar 28 – Apr 3	Role-specific dashboards for employee, manager, admin
Testing & Optimization	Week 11	Apr 4 – Apr 10	Unit testing, integration testing, bug fixes, performance optimization
Deployment & Workflow Setup	Week 12	Apr 11 – Apr 17	Deployment in sandbox; SharePoint workflow triggers and notifications setup
Feedback & Iteration	Week 13	Apr 18 – Apr 24	Stakeholder review, user testing, UI/UX improvement
Finalization & Documentation	Week 14	Apr 25 – May 1	Final code clean-up, internal documentation, deployment guides
Reporting & Presentation	Week 15	May 2 – May 8	Project report prep, final demo to mentors, knowledge transfer



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Outcomes / Results Achieved

- 100% digital workflow for requisitions
- Approval time reduced by 70%
- Zero lost or misplaced forms
- 85% of users rated system as 'Very Useful'
- Rich dataset for future AI/ML



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Challenges Faced / Mitigations

- AD/LDAP sync issues → Overnight batch syncs
- Legacy browser issues → Polyfills & fallback
- Undefined hierarchies → HR workshop
- User resistance → Training & tutorials
- Privacy concerns → RBAC & encryption



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Conclusions and Learnings

Conclusions:

- Web portal improved internal workflows significantly
- Data-driven approvals and scheduling now possible
- Foundation laid for AI/ML integration

Learnings:

- Stakeholder alignment is crucial
- Error-handling avoids workflow blocks
- Modular code eases scaling
- User training drives adoption